Documenting Process Mining Workflows: A Study on Structured Logging in Educational Settings

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Abstract. Process mining workflows often consist of exploratory, data-driven steps that are rarely documented in a structured or reproducible manner. To address this, we drafted a lightweight logging template, aimed at capturing key preprocessing and analysis activities throughout a process mining analysis. The template was piloted in a graduate-level course, where four student groups used it to document their process mining workflows while analyzing real-world event logs. We evaluated the resulting logbooks to assess the completeness and consistency of the documentation. Our analysis shows that while students consistently recorded activities (such as data import, filtering, and process discovery) as well as the tools used and their intended purposes, other fields like parameter settings and outcomes were often documented only at a high level. To deepen our understanding of the workflows and their documentation, we conducted thematic coding of the logs and derived a categorization of business questions, workflow activities and sub-activities, and tool families. Based on these findings, we highlight key areas for improvement in the template design. This study provides early insights into the value and challenges of structured documentation in both educational and applied process mining contexts.

Keywords: Process mining workflow · Workflow documentation · Workflow transparency · Reproducibility in Process Analysis.

1 Introduction

Process mining is a set of techniques used to extract actionable insights from event logs generated by information systems. Its core tasks, including process

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discovery, conformance checking, and enhancement, allow organizations to visualize, monitor, and optimize their real-world processes.

Yet, despite methodological and technical advancements, process mining remains a domain where reproducibility, traceability, and workflow transparency are not sufficiently supported. Most tools lack support for reusable workflows or customization to context-specific attributes, limiting both flexibility and scientific rigor [4]. Recent research has emphasized that process mining is a knowledge-intensive and emergent activity where analysts make iterative and contextual decisions that cannot always be pre-defined [13,2]. Without analytic provenance, i.e., structured records of how data and models are transformed over time, stakeholders cannot retrace the steps that led to specific insights, nor can they evaluate how preprocessing decisions affected outcomes.

This lack of documentation is especially problematic in educational settings, where transparency is essential for reflection and learning. While tools may expose powerful capabilities, students and practitioners require guidance as to why certain filters were applied, how models were discovered, and what impact those decisions had. This type of information is often missing in reported case studies. For instance, a review of submissions to the BPI Challenge 2020 found that only a small fraction of papers reported which techniques were used, making it nearly impossible to trace how process models were generated or interpreted [3]. Similarly, in a recent study on process mining using unstructured data, the authors report that only 6 out of 24 studies provided the necessary source code and datasets, and only 3 studies described their research methods [5]. Automated process discovery techniques are generally applied and assessed in an unstructured way, with researchers using various datasets, evaluation criteria, experimental configurations, and baselines, resulting in findings that are difficult to compare and reproduce [1].

A persistent issue that complicates this further is the quality of event logs. Real-world logs often contain missing or inconsistent data, duplicate events, misaligned timestamps, or heterogeneous case structures [11]. These issues typically require extensive preprocessing before meaningful analysis can begin. Yet, there is little methodological support for selecting the appropriate preprocessing technique for the task at hand [7]. Perhaps because of this, preprocessing is frequently undocumented and executed ad-hoc [6].

To better understand process mining workflows and start collecting workflow documentations, we drafted a structured logging template aimed at supporting transparent documentation of preprocessing and analysis steps in such a process mining workflow. The template aims to capture: (1) Step-level activities, such as log filtering, event abstraction, and process mining tasks; (2) the tool used and its parameters or configurations used to execute the activity (e.g., filter thresholds, case ID settings, discovery miner settings); (3) Purpose and outcomes for each step, including any changes in data size, format, or model characteristics. The template is lightweight and tool-agnostic, requiring manual input but designed to facilitate easy documentation, thereby promoting transparency and reproducibility while supporting reflection.

We piloted the template in a graduate-level process mining course, where four student groups used it to document their workflows while analyzing real-world event logs. Our analysis of the submitted logbooks showed that students recorded core activities—such as data import, filtering, and process discovery—as well as tools and intended purposes. However, fields related to parameter settings and outcomes were often sparsely described or only had high-level descriptions, pointing to the need for clearer guidance or structured support.

Through thematic coding, we further categorized the business questions addressed, the types of workflow activities and sub-activities performed, and the families of tools used. Based on these findings, we identify areas for improvement in the template design and reflect on broader implications for documentation practices in both educational and applied process mining contexts.

2 Related Work

Process Mining Methods and Log Preprocessing Tasks Many efforts have been made to categorize and review process mining tasks, particularly in the area of event log preprocessing. Marin-Castro and Tello-Leal [7] reviewed 70 papers from 2005 to 2020 that explicitly addressed event log cleaning and preprocessing. Van Zelst et al.[12] proposed a detailed taxonomy of event abstraction techniques. Stein Dani et al.[10] highlighted that log extraction consists largely of manual tasks. Liu et al. [6] performed a systematic literature study surveying process mining case studies and identified six high-level preprocessing tasks and twenty-two low-level tasks. Pradhan et al. [8] also performed a systematic literature study, reviewing the pre-analysis activities in process mining projects, and identified 15 pre-analysis activities and concepts, such as data extraction, generation, and cleaning.

While these works contribute valuable taxonomies and insights, few have studied how such categorizations can support the documentation of process mining workflows in practice. Our study builds on this gap by investigating how a lightweight logging template can help document workflow steps.

Process Mining Workflow and Documentation Efforts like RapidProM demonstrate how process mining can be embedded into scientific workflow systems to promote reproducibility.

However, this approach assumes all algorithms to be embedded in a single platform, which contrasts with the heterogeneous tool ecosystems (e.g., Disco, Celonis, PM4Py, Python, ProM), as students and practitioners typically stitch together functionality across multiple environments.

Gamification of such a workflow has been explored to improve data quality and engagement. Particularly, gamification has been applied to improve the quality of process activity labels and to increase the motivation of experts. Sadeghianasl et al. [9] show that a combination of gamification and crowdsourcing can engage process modeling users in data curation. Yet these approaches still face limitations in handling ad-hoc, multi-tool analyses, especially in educational or resource-constrained environments.

Burattin et al. [2] introduce a distributed, privacy- and IP-aware platform for defining, deploying, and executing process mining pipelines with controlled sharing of data and algorithms. While powerful, such systems are heavyweight and best suited for formal, collaborative settings.

In contrast, our work proposes a lightweight, tool-agnostic logging template to bridge the gap between informal notes and heavyweight integrated provenance systems. It aims to support clearer documentation, which helps improve reflection, reproducibility, and transparency, particularly suited for exploratory and educational process mining projects.

3 Logging Template Design

A *process mining workflow* refers to the sequence of steps an analyst follows to extract insights from event logs using process mining techniques to answer a business question. Such workflows are commonly found in process mining case studies, student-led projects, or BPI Challenge reports, and typically involve a mix of technical and analytical tasks. Well-documented workflows promote transparency, traceability, and reproducibility, which are essential for reflection, validation, and knowledge sharing.

To support and encourage the documentation of these workflow steps, we designed a lightweight, structured logging template. The template is intended to help students and researchers document the steps of their analysis, while reflecting on the rationale behind their choices and the outcomes observed.

The template includes the **business question** and the following fields:

- **Order:** Sequential step number indicating the order of operations.
- **Activity:** A brief description of the operation performed (e.g., *Data Import*, *Filtering*, *Discovery*).
- **Tool:** The software, library, or platform used to perform the step (e.g., *PM4Py*, *ProM*, *Disco*).
- **Parameters / Configuration:** Any configuration details or parameter settings applied (e.g., filter thresholds, miner selection).
- **Purpose:** The goal or intention behind performing the step (e.g., *Remove noise*, *Improve fitness*).
- **Outcome / Notes:** Observed results, relevant observations, or comments (e.g., data reduction, model type).

Tables 1 and 2 show two illustrative examples of how the template can be filled in. Each process mining analysis often involves a sequence of exploratory and iterative steps, many of which go undocumented or are only captured very implicitly in project reports. For example, a step such as filtering outliers is often mentioned without specifying the method, tool, or threshold used.

Due to space constraints or reporting priorities, workflow steps are frequently summarized at a high level, omitting essential details such as the tool, the algorithm, or the parameters applied, making it difficult to reproduce or evaluate the analysis. The proposed template offers a lightweight yet structured way to

Table 1: Example 1.

Order	**Activity**	**Tool**	**Parameters/ Config**	**Purpose**	**Outcome/ Notes**
1	Data Import	PM4Py	–	Load XES log	File successfully loaded
2	Pre-processing	Python (pandas)	Removed empty cases	Clean noisy data	Reduced from 10k to 9.2k cases
3	Discovery	ProM (Inductive Miner)	Default settings	Discover process model	Obtained a sound model with loops

Table 2: Example 2.

Order	**Activity**	**Tool**	**Parameters/ Config**	**Purpose**	**Outcome/ Notes**
1	Data Import	Disco	–	Load XES log	File successfully loaded
2	Pre-processing	Disco - variant filter	Top 50%	Remove infrequent cases	Reduced from 9k to 5k cases
3	Discovery	Disco - process map	Default settings, path filter = 0.8	Discover process map	Obtained a map with frequencies

address this gap. It provides a shared format for documenting process mining workflows in a consistent manner. By capturing key metadata (e.g., tools, parameters, purpose, and outcomes), the template encourages users to articulate both the intent and effect of each step. The resulting logbook can be included as an appendix, revisited when deeper review or replication is desired. Structurally, it mirrors the concept of an event log, making it easier for process mining practitioners to adopt and interpret.

4 Methods

We conducted an exploratory study to test out the proposed template. The study was embedded within a graduate-level course on process mining. As part of the course, ten student groups (each consisting of four members) performed a process mining project. Each group selected one real-world event log of interest, out of the three provided (BPIC14, Sepsis, or BPIC19). The assignment was to formulate two to three business questions of interest and design and execute their own analyses using appropriate process mining techniques to address these questions.

To support the documentation of their workflows, students were given the logging template described in Section 3. Completion of the template was voluntary. Four out of the ten groups chose to use the template and provided consent for their data to be included in this study.

The objective of this study is to explore whether and how the template helps students document their process mining workflows. We analyzed the completed logbooks submitted by the four participating groups, each of which documented their end-to-end analysis process using the template.

5 Results

5.1 Quantitative summary

As mentioned, a total of four teams participated in the study, submitting logbooks for 16 individual analyses. The number of logged steps per analysis ranged from 5 to 22 steps, resulting in a combined total of 151 documented steps across all submissions. After initial examination, we also performed thematic coding to label (1) the business questions, (2) activities (using the values in *Activity*, *Purpose*, and *Outcome*), and (3) the tools, to provide a more accurate quantitative summary of the logbooks. In the following, we discuss each of these aspects.

Questions and themes Table 3 presents the 16 business questions formulated by the students, along with the thematic labels we assigned. Following the thematic coding method, we derived the following six themes: process comparison, compliance (e.g., Service-Level Agreements (SLA)), impact/causal analysis, pattern discovery, process prediction, and performance analysis (e.g., bottleneck detection).

We observed two distinct approaches to structuring and reporting the questions and corresponding analyses: (1) beginning with a simple question and then breaking it down into multiple sub-questions, resulting in shorter and more focused analyses (see Team A), and (2) formulating a more complex question that guided multiple consecutive analytical steps within a single, cohesive analysis (see the two questions raised by Team C). This distinction helps explain the variation in the number of steps documented per question.

Among the six types, `process comparison` was a frequently listed type of question/analysis, appearing in 5 out of 16 questions (31%). These questions primarily are phrased about the differences across groups, such as priority levels, patient age groups, or closure codes. All four teams included at least one process comparison in their project, suggesting that this type of analysis is broadly applicable and of interest. However, we observed that while such questions can focus on various aspects (e.g., process behavior, variant distribution, performance differences), the available tools used in the course offered limited support for analyzing differences in process behavior. In these cases, students often relied on and reported the basic “log statistics” or performed visual comparisons of visualizations, which limited both the depth of analysis and its transparency and reproducibility.

`Compliance analysis` (e.g., SLA adherence), was also mentioned in 5 questions. This type of analysis was combined with an impact/causal analysis (e.g., “effect of work distribution on compliance”).

Table 3: Categorization of Questions by Analysis Type

Team	**Steps**	**Question**	**Process Comparison**	**Compliance (SLA)**	**Impact/causal**	**Pattern discovery**	**Process prediction**	**Performance**
A	5	Do high-priority incidents follow different process paths compared to low-priority ones?	1					
A	4	To what extent is the incident management process meeting the 48-hour SLA target, and how do workload distribution impact compliance and operational efficiency?		1	2			
A	10	Pattern identification and predicting active cases likely to become stuck.				1	2	
A	5	Where and why bottlenecks occur in BPIC 2014 incident management process?						1
A	5	Which subprocesses and products are affected more?	1					
A	4	Do reassignments contribute to the bottlenecks?			1			
A	4	How much overall resolution time is lost while incidents wait in the “Assign to 2nd Line Operator” queue...?						1
A	4	At what point does the number of ticket re-assignments (‘ping-pong’) start to inflate resolution time disproportionately?			2	1		
A	4	In how many cases is Impact & Urgency not set within 30 minutes of ‘Open’, and what share of those non-compliant tickets miss the SLA or get re-opened?		1				
B	11	Does the common treatment process differ across multiple patient age groups?	1					
B	14	How do the process paths and the activities therein differ between one-off and returning patients	1					
B	11	Does timely treatment actually lead to a positive outcome?			1			
C	21	In which category of items does non-compliance most severely disrupt the expected process flow?		1	2			
C	22	To what extent do specific handovers between user roles or systems correlate with increased case durations in the purchase-to-pay process?			3	1		2
D	7	Does the process meet the demand to resolve processes with a higher priority faster?		1				
D	20	How do incident handling processes differ across Closure Codes, and what causes longer handling durations for specific Closure Codes?	1		3			2

`Performance analysis` appeared in 4 questions. These questions typically mention “bottlenecks”, “resolution time”, “case duration”, and “handling duration”. Examining the analysis performed, they typically also use log statistics, process maps (enhanced with performance information), and visualizations to identify process delays or inefficiencies.

`Impact/causal analysis`, often targeting a single feature, was applied in 7 questions (44%) and was very frequently conducted in combination with other themes of questions or as a follow-up analysis step. These questions commonly aimed to explain non-compliance behavior, rework (e.g., reassignment loops), or performance issues. This suggests a strong interest among students not only in identifying what occurred, but also in understanding why it happened.

Activity, Purpose, and Outcome Across the 151 logged steps, students reported around 80 distinct activities. We also observed two general styles of reporting the workflow steps: (1) high-level activities based on template examples (e.g., data import, filtering), and (2) more customized, fine-grained descriptions.

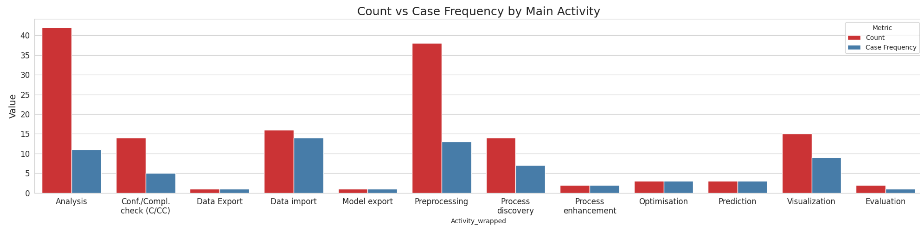

Fig. 1: Frequency of logged activity types across all teams.

After merging duplicates (e.g., "pre-processing" and "preprocessing"), the most frequently reported workflow activities were `discovery` (27 steps), `data import` (18), `preprocessing` (15), and `data export` (5). Several other categories were reported only two or three times, while 67 activities appeared only once, reflecting considerable variation in how students documented their workflows. In some cases, activity labels did not align well with the described purpose or outcome. For example, a step labeled "Discovery" might refer to plotting metrics rather than model generation.

To address this, we re-coded the entries and grouped them into 12 main activity categories. Among these, `Analysis` (e.g., comparing, calculating, identifying) and `Preprocessing` (e.g., filtering, enriching, bucketing) were the most prominent. Analysis appeared 42 times across 11 of 16 cases; Preprocessing, 38 times across 13 cases—indicating both were central to the students' workflows, with Preprocessing being more consistently applied.

Other common categories included `Data import`, `Visualization`, `Process discovery`, and `conformance/compliance checking (C/CC)`, while less frequent ones like `Optimization` and `Prediction` appeared only in specific contexts. Figure 2 shows sub-activity distributions within each main category. In Analysis, frequent sub-activities included Resource analysis, Performance analysis, and Comparison; in Preprocessing, Filtering (15), Enriching (6), and Bucketing (5). `Bucketing`, which is often described as creating sublogs using attributes or conditions, was very frequently used together with comparison. For example, "a sublog for each priority" was created and they are compared.

Tool Usage We categorized the tools reported in the workflow logs into four main families: Python, PM4Py, Disco, and Other, with further division into 19 sub-families based on specific techniques or modules. As shown in Figure 3, Python (75 steps) and PM4Py (57) were used most frequently, followed by Disco (16).

Within the Python family, students primarily used the tool related to data handling (18 steps, such as pandas, numpy, etc.), custom logic (17), and visualization (12), with fewer mentions of statistical analysis and other tasks (5 each). Notably, 18 Python-related entries lacked specific sub-family details. Similarly, nearly half of the PM4Py entries (29) did not specify the exact module or functionality used. Among those that did, filtering (9), Inductive Miner (5), conformance checking (4), and log statistics (4) were most common, alongside

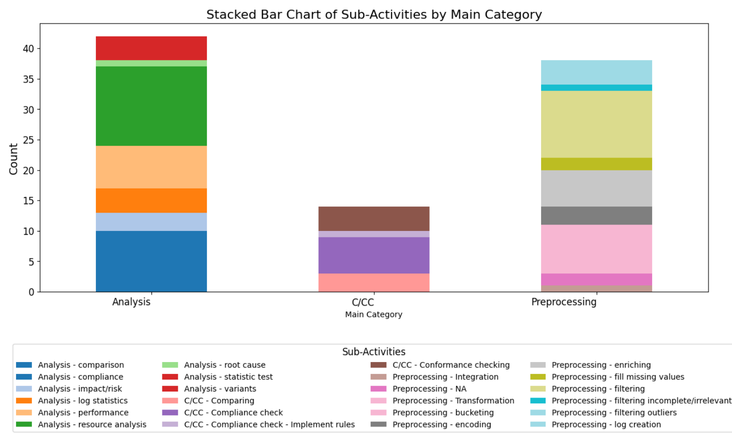

Fig. 2: Frequency of sub-activity types across all teams.

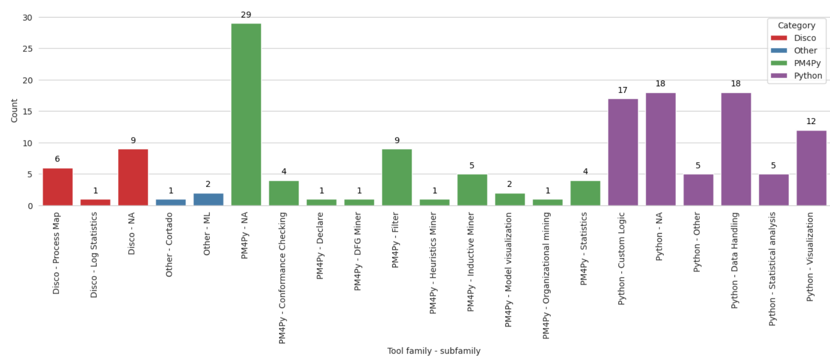

Fig. 3: Frequency of logged tools (classified as reported in Tool Usage). NA means that the second level of the tool is not reported.

occasional use of model visualization and other techniques like Declare and DFG Miner. For Disco, the process map view was the most frequently cited feature (6 steps), with one mention of statistics and 9 unspecified entries. The "Other" category included only three mentions: two related to machine learning tools and one to Cortado.

Overall, while students did documented the primary tools, many did not specify the exact functionality used. When specified, Python-based data handling, PM4Py filtering, or discovery techniques were the most frequently documented.

Template Field Utilization The columns *Activity*, *Tool*, and *Purpose* are filled in for all 151 steps. However, the *Parameters / Config* and *Outcome / Notes* fields were sometimes incomplete or varied greatly in detail.

The *Parameters / Config* field was left blank or marked with a dash in 19 entries (13%), often in steps related to data import or custom preprocessing. Even in cases where parameterization was expected, such as filtering or discovery, specific values were often not documented. Only about 10 entries included concrete configurations, such as filtering conditions, tool-specific settings, or miner parameters (e.g., "*max_ edges = 50, 100, 500)*". The diversity in terminology and structure suggests challenges in standardizing how parameters are reported.

Details in the *Outcome / Notes* field ranged widely. Some students recorded quantitative results (e.g., "*ROC-AUC = 0.826*", "*Reduced from 1050 to 782 cases*"), while most provided high-level or qualitative observations (e.g., "*Identified compliant/non-compliant cases*", "*Network graphs showing role relationships*"). While these summaries offered insight into the analytical direction, they often lacked the detail required for reproducibility.

The high number of vague or omitted entries highlights the need for better guidance on documenting parameters and outcomes. Though the Outcome field helped reveal students' interpretations, more structure and example-driven prompts could enhance clarity and support reproducibility.

5.2 Discussion, Lessons Learned and Future Improvements

Our analysis of the students' workflow logs revealed several important lessons and recurring challenges in documenting both standard and exploratory steps in process mining analyses.

For simple and well-known tasks such as data import, log filtering, process map discovery, and basic visualizations, the logging template supported documenting the steps effectively. These activities are commonly encountered in most process mining analyses and are typically well-supported by tools like PM4Py and Disco. Students were able to report these steps clearly and consistently, reflecting both familiarity with the tasks and alignment with available tooling.

However, challenges emerged when students engaged in more exploratory or customized analyses, such as creating new variables, integrating external data, or scripting logic in Python. While the template still supported documentation, entries were often less structured. In many of these cases, either the activity was inaccurately documented, or the rationale and parameter settings were described only at a high level. This suggests a need to improve the template's support also for non-standard workflows, including providing more flexible descriptors and prompts for documenting rationale, assumptions, and custom logic.

Shared vocabulary, ontology, or repository A key challenge identified was the lack of predefined categories or shared vocabulary for describing analysis activities. This led to highly diverse and sometimes inconsistent documentation. To address this, we recommend developing a shared ontology or repository, or aligning the template with an established process mining methodology, that categorizes common activities (e.g., import, preprocessing, discovery, conformance checking) along with finer-grained subcategories (e.g., filtering, abstraction, enrichment under preprocessing). This ontology should also link tools and techniques to each activity and provide a template to document commonly used parameters and expected outcomes. Such an ontology or repository could help standardize not only the activity labeling but also improve comparability across different workflows.

Standardization Standardization is also important in documenting parameters and results. In many cases, students did not record the exact parameter values used in their filters or discovery algorithms, which makes it difficult to reproduce or evaluate their findings. Encouraging more detailed reporting of parameter settings, expected outcomes, and rationale could significantly improve the transparency and reusability of logged workflows. This may be a call to the PM community tool developers, which should allow for printing a standardized message specifying the parameter/configuration used that can be readily copied to such a logging report.

Automated tool support With a clearer understanding of common process mining workflow steps and associated tools, there is potential to develop an interactive tool to support users in completing the logging template more consistently and accurately. Such a tool could guide users step-by-step and include built-in consistency checks. For example, selecting `data import` could trigger a list of relevant tools (e.g., Disco, ProM, PM4Py), with incompatible options (e.g., Inductive Miner) disabled. If Disco and a CSV file are selected, the tool could prompt for required parameters like case ID and activity columns. Such guided input would improve completeness and standardize documentation.

Concurrent or iterative analysis Some cases revealed structural limitations of the current template. In one analysis, students applied the same sequence of steps to two filtered event logs representing different patient groups. They noted that steps were repeated "for both dataframes (one-off and returning patients)," but the template lacked a clear way to represent parallel workflows. Similarly, another group referenced "separate logs per group" and "common variants per age group." These examples suggest the need for future versions of the template to support parallel branches or concurrent sub-workflows, potentially through grouping mechanisms or step references.

In summary, while the template supports workflow documentation, inconsistent parameter and outcome reporting highlight the need for clearer guidance and standardized terminology.

Limitation Our study was limited to a pilot with four groups of students on a single graduate-level course. The questions and the tools were limited to the content covered by the students' course, and this meant that other tools, for example, BupaR, were not covered. In future work, we aim to further improve the template and repeat the study in different settings.

6 Conclusion

This exploratory study examined the feasibility of a lightweight logging template designed to support easy documentation of process mining workflow steps, in order to improve their transparency and reproducibility. We analyzed 151 workflow steps from 16 analyses across four student teams. The template effectively supported documentation of activities, tools, and purposes, with preprocessing, discovery, comparison, and performance analysis being the most common themes. While the *Activity*, *Tool*, and *Purpose* fields were consistently filled, the *Parameters / Config* and *Outcome / Notes* were often high-level or sometimes incomplete, limiting reproducibility, especially for customized analyses.

These findings highlight the template's value as a starting point, but also point to the need for clearer guidance, examples, and tool-supported logging to improve completeness and depth. Overall, structured logging shows promise as a methodological aid in process mining education and research.

References

1. Augusto, A., Conforti, R., Dumas, M., Rosa, M.L., Maggi, F.M., Marrella, A., Mecella, M., Soo, A.: Automated discovery of process models from event logs: Review and benchmark. IEEE Trans. Knowl. Data Eng. **31**(4), 686–705 (2019)
2. Burattin, A., Kindler, E., Dyhre, N., Vestrup, S., Zerbato, F., Weber, B.: Process mining pipelines with controlled sharing of data and algorithms. In: International Conference on Business Process Management. pp. 357–369. Springer (2024)
3. Hoogendoorn, T., Arachchige, J.J., Bukhsh, F.A.: Survey of explainability within process mining: A case study of bpi challenge 2020. In: 2023 International Conference on Frontiers of Information Technology (FIT). pp. 43–48. IEEE (2023)
4. Janssenswillen, G., Depaire, B., Swennen, M., Jans, M., Vanhoof, K.: bupar: Enabling reproducible business process analysis. Knowledge-Based Systems **163**, 927–930 (2019)
5. König, F., Egger, A., Kratsch, W., Röglinger, M., Wördehoff, N.: Unstructured data in process mining: A systematic literature review. ACM Transactions on Management Information Systems **16**(3), 1–34 (2025)
6. Liu, Y., Dani, V.S., Beerepoot, I., Lu, X.: Turning logs into lumber: Preprocessing tasks in process mining. In: ICPM Workshops. LNBIP, vol. 503, pp. 98–109. Springer (2023)
7. Marin-Castro, H.M., Tello-Leal, E.: Event Log Preprocessing for Process Mining: A Review. Applied Sciences **11**(22), 10556 (nov 2021)
8. Pradhan, S.K., Jans, M., Martin, N.: Getting the data in shape for your process mining analysis: An in-depth analysis of the pre-analysis stage. ACM Comput. Surv. **57**(6), 159:1–159:37 (2025)
9. Sadeghianasl, S., Ter, A., Suriadi, S., Turkay, S.: Collaborative and interactive detection and repair of activity labels in process event logs (10 2020)
10. Stein Dani, V., Leopold, H., van der Werf, J.M.E.M., Lu, X., Beerepoot, I., Koorn, J.J., Reijers, H.A.: Towards understanding the role of the human in event log extraction. In: BPM Workshops. pp. 86–98. Springer International (2022)
11. Suriadi, S., Andrews, R., ter Hofstede, A.H., Wynn, M.T.: Event log imperfection patterns for process mining: Towards a systematic approach to cleaning event logs. Information systems **64**, 132–150 (2017)
12. van Zelst, S.J., Mannhardt, F., de Leoni, M., Koschmider, A.: Event Abstraction in Process Mining: Literature Review and Taxonomy. Granular Computing **6**(3), 719–736 (2021)
13. Zerbato, F., Burattin, A., Völzer, H., Becker, P.N., Boscaini, E., Weber, B.: Supporting provenance and data awareness in exploratory process mining. In: CAiSE. Lecture Notes in Computer Science, vol. 13901, pp. 454–470. Springer (2023)